Vince Németh

Applied Data Science & AI Student · Full-Stack ML Engineering · Real-World Industry Projects

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PROFESSIONAL SUMMARY

Applied Data Science & AI student and hands-on ML engineer with real industry delivery. I work full-stack across the ML lifecycle: training custom deep-learning and Transformer models, then containerising, deploying, and monitoring them across local, on-premise, and cloud (Azure ML) environments. I'm comfortable owning the foundations that make ML production-grade: data layers, model-lineage and audit tooling, CI/CD, and self-hosted infrastructure. Recent work spans a production MLOps platform, RAG for engineering docs, autonomous CV + robotics phenotyping, and quantitative NLP research.

CORE COMPETENCIES

Languages & Backend: Python, SQL, FastAPI, PostgreSQL, SQLAlchemy, Alembic, PySide6 (Qt)

AI & Deep Learning: PyTorch, TensorFlow, Transformers (RoBERTa), CNNs, ResUNet, ResNet50, LSTM, MediaPipe

NLP & LLMs: LLMs, RAG Systems, ChromaDB, Custom Embeddings, Sentiment / Emotion Classification, Speech-to-Text, Machine Translation

MLOps & Cloud: Azure ML, MLflow, Docker, Docker Compose, Portainer, CI/CD (GitHub Actions), Model Versioning & Lineage, Automated Retraining, Experiment Tracking

Robotics & Control: PID Control, Reinforcement Learning, Opentrons OT-2, Hardware Integration

DevOps & Infrastructure: Linux / Bash, Git / GitHub, Traefik, Self-Hosting

PROFESSIONAL EXPERIENCE

Junior Software Developer — Jul 2025 – Aug 2025

Valeo

- Built tools to analyse large volumes of test data (1M+ records across 100+ product SKUs) with automated anomaly detection, trend analysis, visualisation, and reporting.
- Developed a Cpk process-capability analysis application adopted by 2 engineering teams, replacing manual spreadsheet workflows for production-line evaluation.
- Contributed to project feasibility assessment and digitisation strategy.

Supplier Quality Assurance Assistant — Jun 2024 – Jul 2024

Valeo

- Supported 3 supplier audits, identifying quality deviations across production batches.
- Worked with quality-control analytics and production-data workflows; gained insight into the quality-control logic used in industrial automation.

Freelance Software Engineer — 2023 – Present

Independent

- **CineSlice**: desktop video frame extractor built on a multi-threaded FFmpeg pipeline (Python, PySide6).
- **Infrastructure Monitor**: self-hosted alerting and monitoring system with Docker, Traefik, and SEO validation.

FEATURED PROJECTS

NPEC: Autonomous Root Phenotyping & Production ML Platform — Breda UAS

Netherlands Plant Eco-phenotyping Centre (NPEC) · Academic Industry Project

A two-phase engagement for high-throughput plant phenotyping: first an autonomous computer-vision and robotics pipeline, then a team-built platform that takes the resulting model into production.

- **Phenotyping pipeline:** built a Deep ResUNet model for Arabidopsis root segmentation and implemented PID and reinforcement-learning control for the Opentrons OT-2, integrated into a complete system that detects root tips and performs precise inoculation.
- **Production platform (team):** a full-stack MLOps platform serving the root-detection model in production: an authenticated web frontend and FastAPI backend running the segmentation pipeline, deployable unchanged across local (Docker Compose), on-premise (Portainer), and Azure ML.
- **My contribution (data layer):** designed and built the entire PostgreSQL data layer, a nine-table SQLAlchemy schema with Alembic migrations and a Dockerised migration runner kept consistent across all environments.

- **My contribution (model lineage):** built a lineage & audit system that registers every model against its git commit, logs each prediction to an append-only audit trail, and traces any prediction back to the exact model, dataset, and commit.

Tech: Python · PyTorch · ResUNet · PID Control · Reinforcement Learning · Opentrons OT-2 · PostgreSQL · SQLAlchemy · Alembic · FastAPI · Docker · Portainer · Azure ML · MLflow

Tobroco-Giant: AI Documentation Assistant — Breda UAS

Tobroco-Giant · Academic Industry Project

Designed and deployed a semantic-search and Retrieval-Augmented Generation (RAG) system for engineering manuals, enabling intelligent document querying and AI-assisted authoring.

- Built a FastAPI backend with a ChromaDB vector database for semantic retrieval.
- Delivered a working prototype to stakeholders; the project is advancing to a professional deployment phase.

Tech: Python · FastAPI · ChromaDB · RAG · LLMs · Custom Embeddings

Context-Aware Translation for Emotion Classification — Breda UAS

Quantitative Research Study · Research Methodology

Studied whether document-level context improves emotional nuance in machine translation. Benchmarked 3 MT engines (Opus-MT, NLLB-200, Llama 3.1 70B) across 4 context windows on 1,045 Spanish dialogue utterances, evaluated with McNemar, Cochran's Q, Friedman, and bootstrap confidence intervals. Key finding: the benefit of context is architecture-dependent: sentence-pair models collapse (−54% F1) while only the instruction-tuned LLM stays stable and slightly improves.

Tech: Python · Transformers · Opus-MT · NLLB-200 · Llama 3.1 · Statistical Hypothesis Testing

ADDITIONAL PROJECTS

Latent Space: Through the Eyes of the Algorithm, an interactive art installation turning human gaze into AI-interpreted artwork: eye movements tracked via MediaPipe, rendered as abstract strokes, classified with ResNet50, and interpreted through custom LLM prompts into a thermal-print keepsake.

Hungarian Video Emotion Classification Pipeline, an end-to-end NLP pipeline: audio extraction to speech-to-text (AssemblyAI), translation (Helsinki-NLP OPUS-MT), then 7-class emotion classification using RoBERTa embeddings fed into a trained LSTM classifier.

Digital Advisor (AI Stock Forecasting): real-time stock-movement forecasts using LSTM models, backed by PostgreSQL and served through a FastAPI dashboard.

Automated Vehicle Damage Classifier: deep-learning image classifier for automated car-damage assessment, detecting cracks and dents, with room to expand to scratches, broken glass, and lamp damage.

TrueNAS Server: ZFS-backed NAS with Dockerised services and automated backups.

EDUCATION

BSc Applied Data Science & Artificial Intelligence

Breda University of Applied Sciences · 2024 – Expected 2027–2028 · Focus: Deep Learning, NLP, Computer Vision, Robotics, MLOps, Research Methodology.

French Bilingual Faculty · Vetési Albert Gimnázium · 2019–2024 · Advanced Mathematics & Computer Science

CERTIFICATIONS & LANGUAGES

Cambridge C2 Proficiency (CAE): Score 190 · **English**: C2 · **Hungarian**: Native · **French**: Intermediate